

See the section “Indexing a Two-dimensional Array” in <https://www.pluralsight.com/guides/working-numpy-arrays-indexing-slicing-guide>

See the section “Slicing a Two-dimensional Array” in <https://www.pluralsight.com/guides/working-numpy-arrays-indexing-slicing-guide>

Lab4_5: Additional Exercises

#Ex16

Create a 100×100 matrix with random integers $\in [1, 9]$, and then making all elements that are larger than 5 to be set as 1, and all elements that are less than 5 is set to 0, elements equal to 5 is unchanged.

- For ex., input matrix = $\begin{bmatrix} 2 & 5 & 9 \\ 3 & 5 & 8 \\ 4 & 4 & 7 \end{bmatrix} \rightarrow \text{output} = \begin{bmatrix} 0 & 5 & 1 \\ 0 & 5 & 1 \\ 0 & 0 & 1 \end{bmatrix}$

#Ex17

Create a 100×100 matrix with random integers $\in [1, 9]$, and calculate the sum of the main diagonal elements of the matrix.

- For ex., input matrix = $\begin{bmatrix} 2 & 5 & 9 \\ 3 & 5 & 8 \\ 4 & 4 & 7 \end{bmatrix} \rightarrow \text{output} = 14$

#Ex18

Create a 15×15 magic square matrix.

Hint: A Magic Square is a $n \times n$ matrix of the distinct elements from 1 to n^2 where the sum of any row, column, or diagonal is always equal to the same number.

- For ex., a Magic Square of size 3 = $\begin{bmatrix} 2 & 7 & 6 \\ 9 & 5 & 1 \\ 4 & 3 & 8 \end{bmatrix}$

#Ex19

Create a 100×100 matrix with 1 on the borders and 0 inside.

- For ex., $\begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \\ 1 & 1 & 1 & 1 \end{bmatrix}$

#Ex20

Create a 100×100 matrix with random integers $\in [1, 9]$, and add a border (filled with 0's) around the matrix.

- For ex., input matrix = $\begin{bmatrix} 2 & 5 & 9 \\ 3 & 5 & 8 \\ 4 & 4 & 7 \end{bmatrix} \rightarrow \text{output} = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 2 & 5 & 9 & 0 \\ 0 & 3 & 5 & 8 & 0 \\ 0 & 4 & 4 & 7 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

#Ex21

Create a 100×100 matrix with random integers $\in [1, 9]$, and sort each row of the matrix in ascending order.

$$- \text{ For ex., input matrix} = \begin{bmatrix} 5 & 2 & 9 \\ 8 & 5 & 3 \\ 4 & 4 & 7 \end{bmatrix} \rightarrow \text{output} = \begin{bmatrix} 2 & 5 & 9 \\ 3 & 5 & 8 \\ 4 & 4 & 7 \end{bmatrix}$$

#Ex22

Create a 100×100 matrix with random integers $\in [1, 100]$, and get the sum of all the multiples of 3 or 5 in the matrix.

$$- \text{ For ex., input matrix} = \begin{bmatrix} 15 & 2 & 9 \\ 8 & 5 & 3 \\ 4 & 4 & 7 \end{bmatrix} \rightarrow \text{output} = 15 + 5 + 3 + 9 = 32$$

#Ex23

Create a 100×100 matrix with random integers $\in [1, 1000]$, and get the five smallest elements in the matrix.

$$- \text{ For ex., input matrix} = \begin{bmatrix} 15 & 2 & 9 \\ 8 & 5 & 3 \\ 4 & 4 & 7 \end{bmatrix} \rightarrow \text{output} = [2, 3, 4, 5, 7]$$

END

Feel tired, relax here: <https://www.youtube.com/watch?v=nGRKl1Owing>